ReAct: Synergizing Reasoning and Acting in Language Models

by Shunyu Yao, Jeffrey Zhao et al. (2023)

Article's overview,
results reproduction and new findings

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The ReAct Paradigm

General setup:

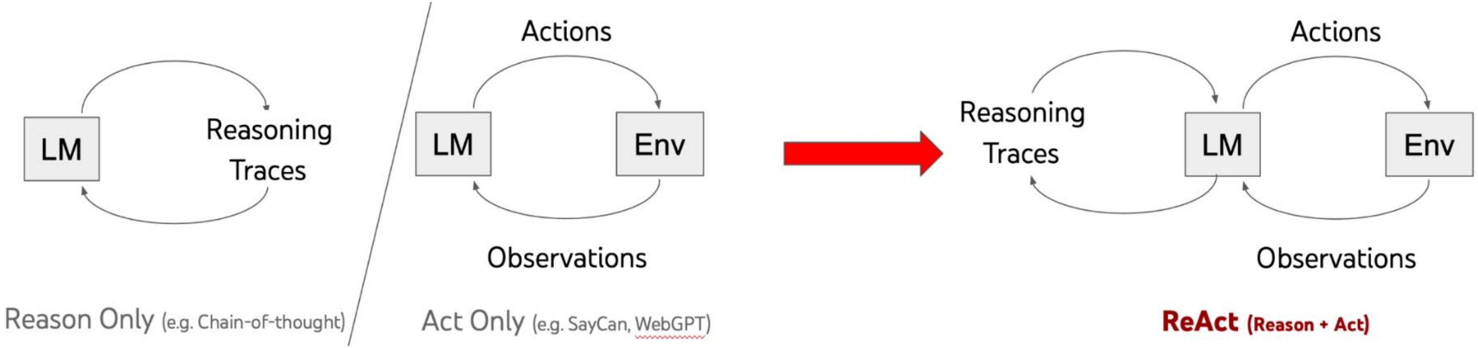
Observation o_t ∈ O => action a_t ∈ A following some policy $\pi(a_t|c_t)$, where $c_t = (o_1, a_1, \cdots, o_{t-1}, a_{t-1}, o_t)$ is the context to the agent.

ReAct's setup:

Augmented agent's action space to A* = A ∪ L, where L is the space of language.

action a_t∈ L in the language space (thought) => no observation feedback.

a_t => reasoning over the current context c_t => update the context $c_{t+1} = (c_t, a_t)$



Methodology

- **Prompt Design**
  Few-shot examples demonstrate ReAct's thought-action format.
- **Action Space**
  Task-specific actions (e.g., search, answer) are predefined.
- **Environment**
  External systems execute actions and return observations.
- **Execution**
  Model alternates thoughts and actions until task completion.

ReAct

```
Thought 1: I need to search Apple Remote and find the program
it was originally designed to interact with.
Act 1: Search[Apple Remote]
Obs 1: The Apple Remote is a remote control introduced in
October 2005 by Apple … originally designed to control the
Front Row media center program …

Thought 2: Apple Remote was originally designed to control the
Front Row media center program. I need to search Front Row
next and find what other device can control it.
Act 2: Search[Front Row]
Obs 2: Could not find [Front Row]. Similar: ['Front Row Seat
to Earth', 'Front Row Motorsports','Front Row (software)', …]

Thought 3: Front Row is not found. I need to search Front Row
(software) .
Act 3: Search[Front Row (software)]
Obs 3: Front Row is a discontinued media center software …

Thought 4: Front Row (software) is controlled by an Apple
Remote or the keyboard function keys. So the answer is
keyboard function keys.
Act 4: Finish[keyboard function keys]
```

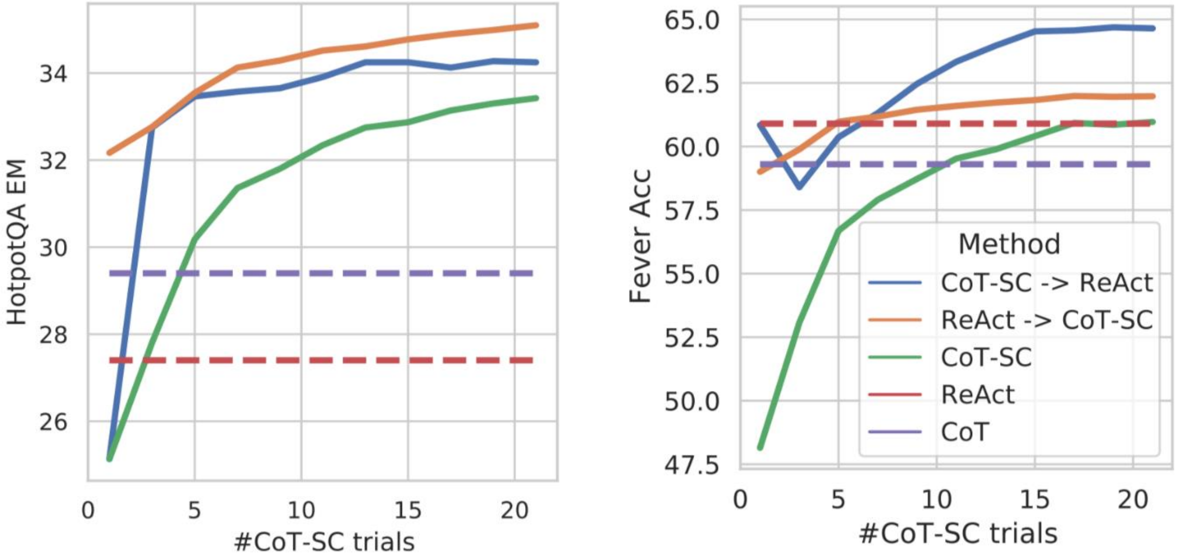
Benchmark Tasks

- **Question Answering (HotpotQA)**: A multi-hop QA dataset requiring reasoning across multiple pieces of information
- **Fact Verification (FEVER)**: Determining whether claims are supported, refuted, or unverifiable based on evidence
- **Text-based Game (ALFWorld)**: Navigating and completing household tasks in a text-based virtual environment (organized text environment)
- **Web Shopping (WebShop)**: Finding products on a simulated e-commerce website based on natural language instructions (noisy environment, unstructured texts)

The authors used frozen **PaLM-540B** (prompted with few-shot in-context examples)

Key Results: HotpotQA, FEVER

CoT (CoT-SC) vs ReAct



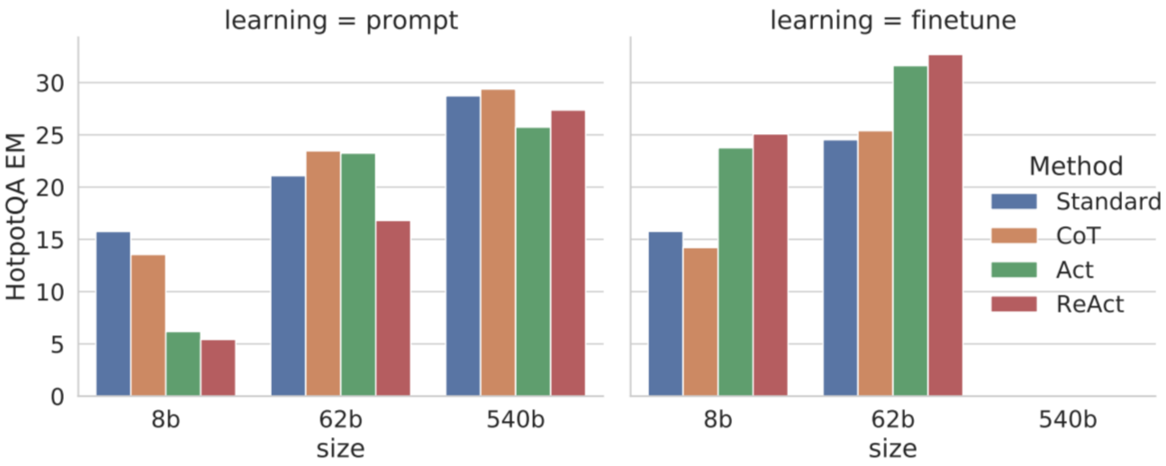
Prompt Method[a]	HotpotQA (EM)	Fever (Acc)
Standard	28.7	57.1
CoT (Wei et al., 2022)	29.4	56.3
CoT−SC (Wang et al., 2022a)	33.4	60.4
Act	25.7	58.9
ReAct	27.4	60.9
CoT−SC → ReAct	34.2	**64.6**
ReAct→ CoT−SC	**35.1**	62.0
Supervised SoTA[b]	67.5	89.5

Table 1: PaLM-540B prompting results on HotpotQA and Fever.

* CoT-SC (self-consistent CoT) - sampling 21 CoT trajectories with decoding temperature 0.7 during inference and adopting the majority answer

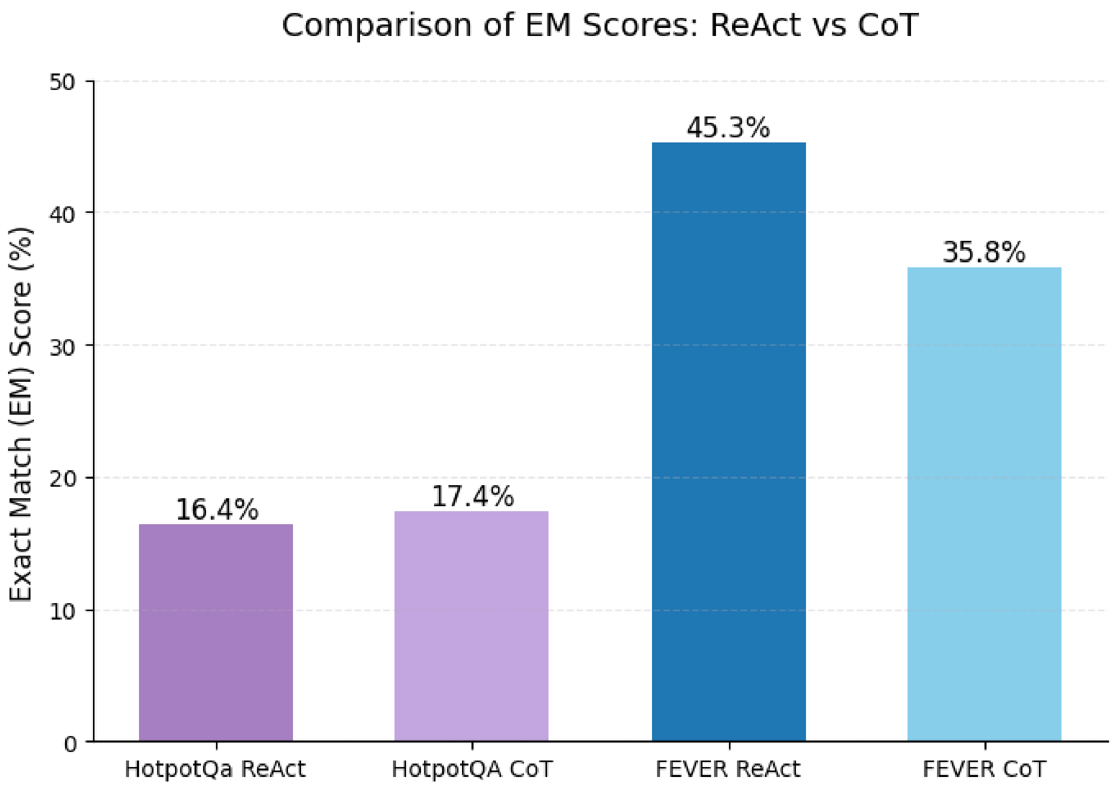
Key Results: HotpotQA, FEVER

*Finetuning PaLM-8b, -62b



Scaling results for prompting and finetuning on HotPotQA with `ReAct` (ours) and baselines.

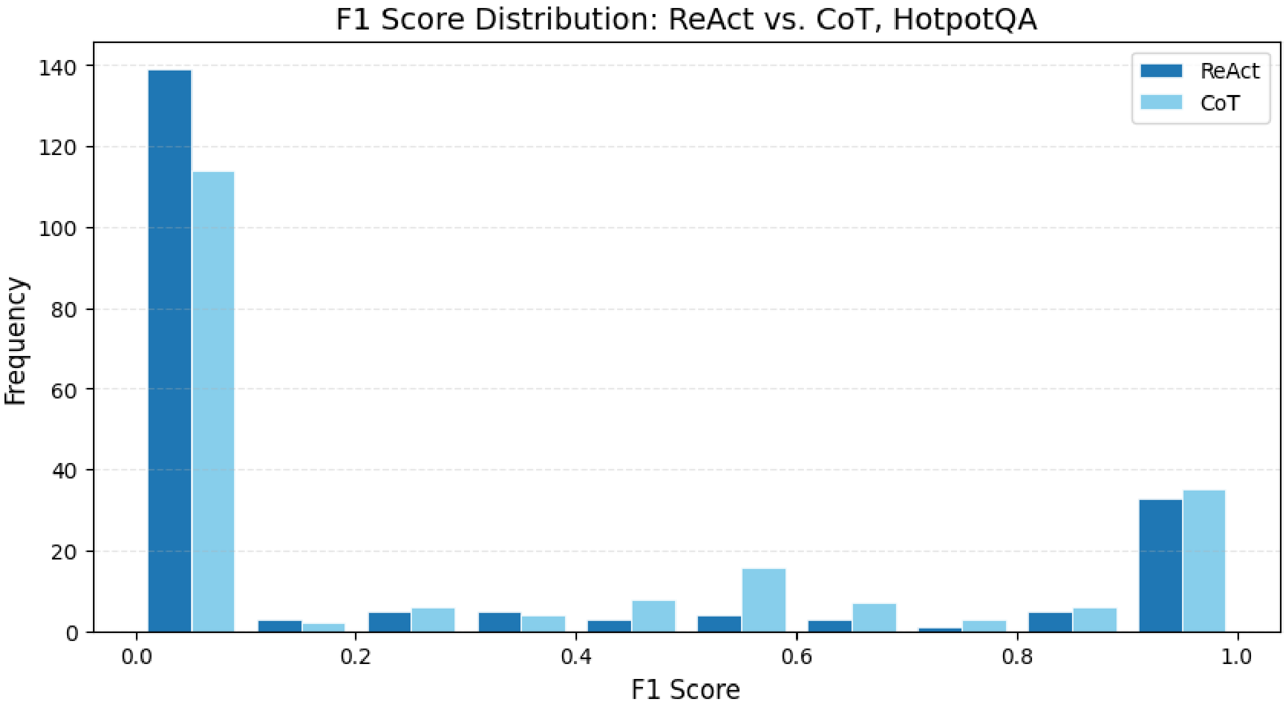
Reproduction of experiments



Setup:

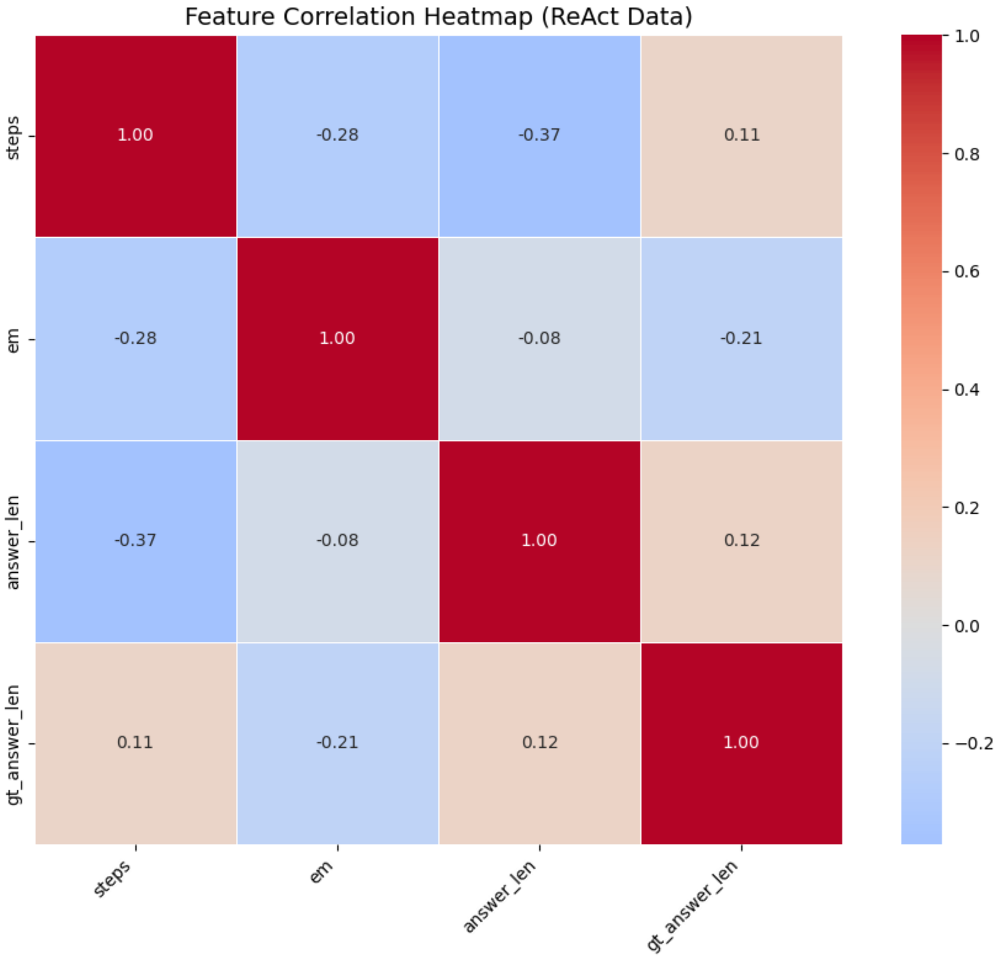
- Qwen-8b (quantized, temperature 0.1)
- ReAct framework (initial author's version)
- CoT (Qwen-8b prompted to think step-by-step)
- 200 questions/claims
- Learning = prompt

HotpotQA Benchmark



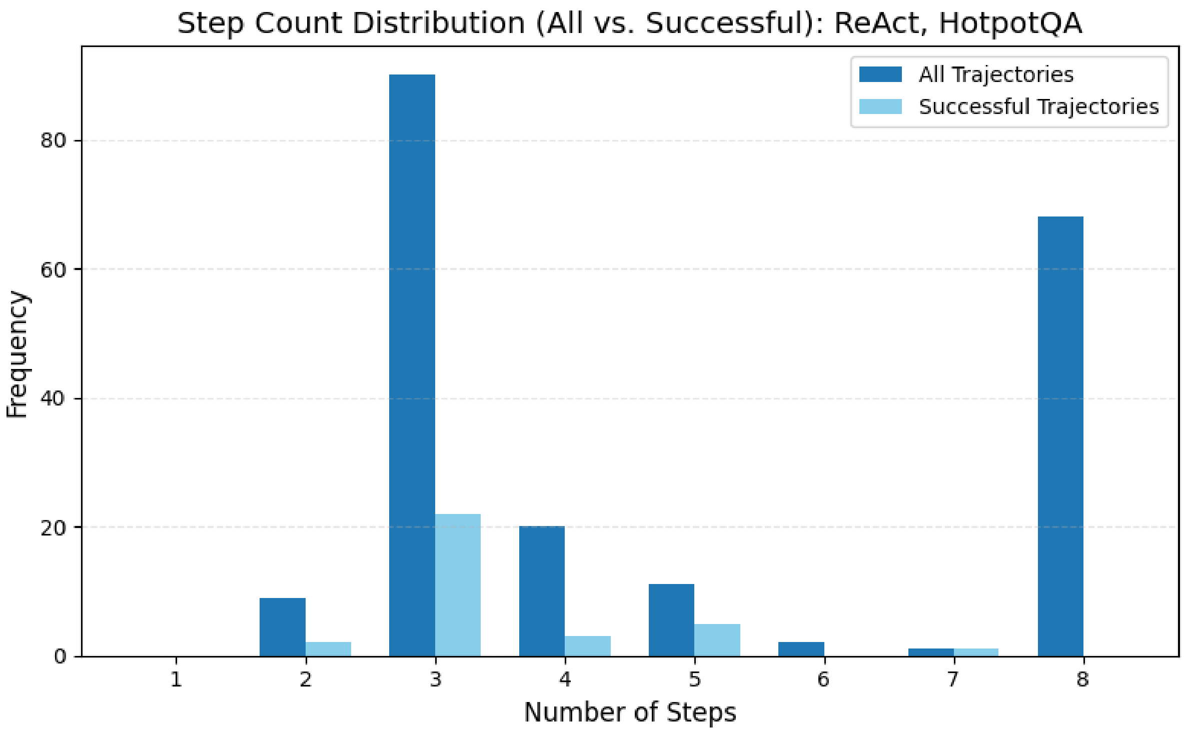
- CoT outperforms HotpotQA
- Label ambiguity

HotpotQA Benchmark



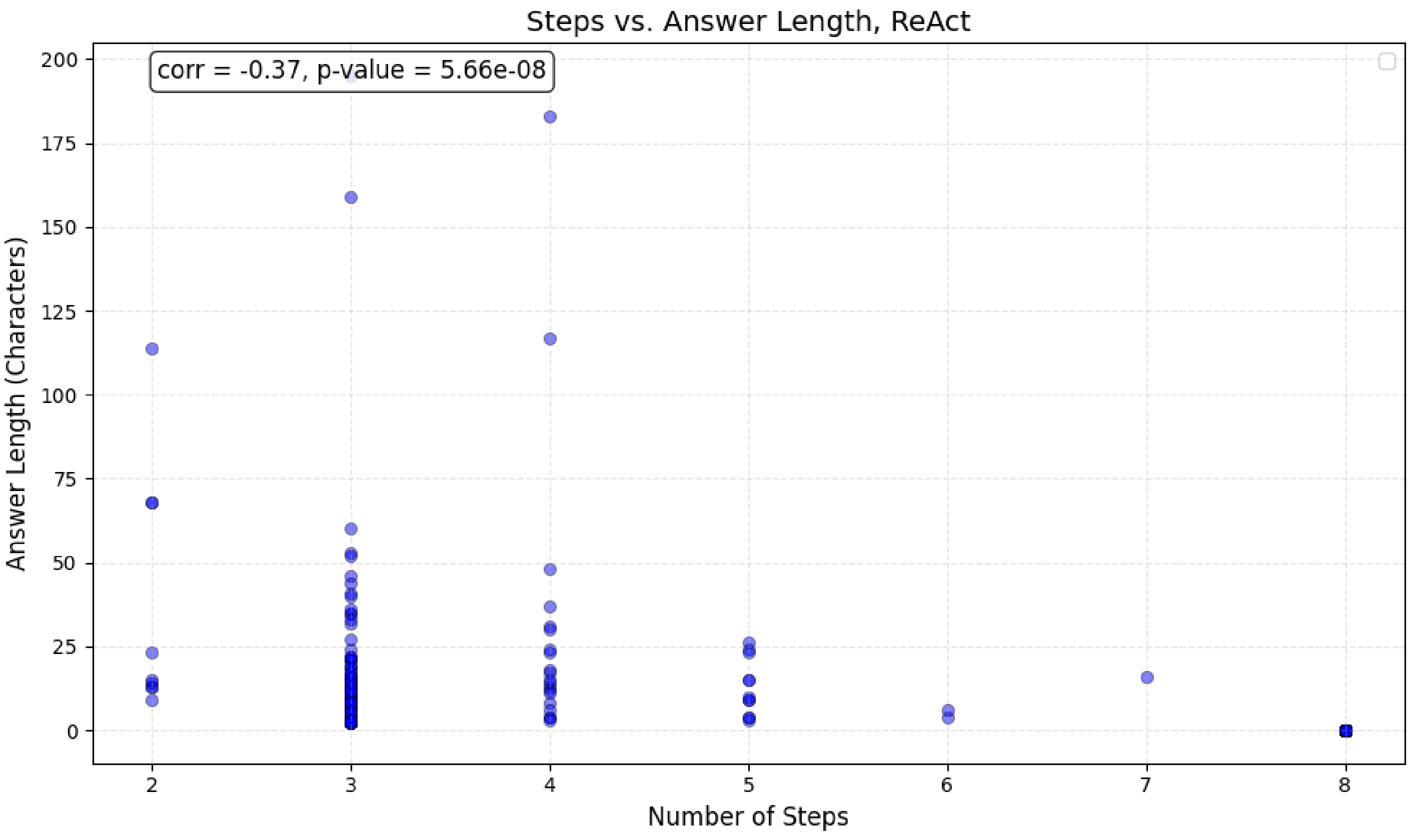
- Correlation between number of steps and EM / answer's length

HotpotQA Benchmark



- Optimal number of steps < 5
- Entering infinite loops (reaching maximum 8 steps) leads to empty string as an answer

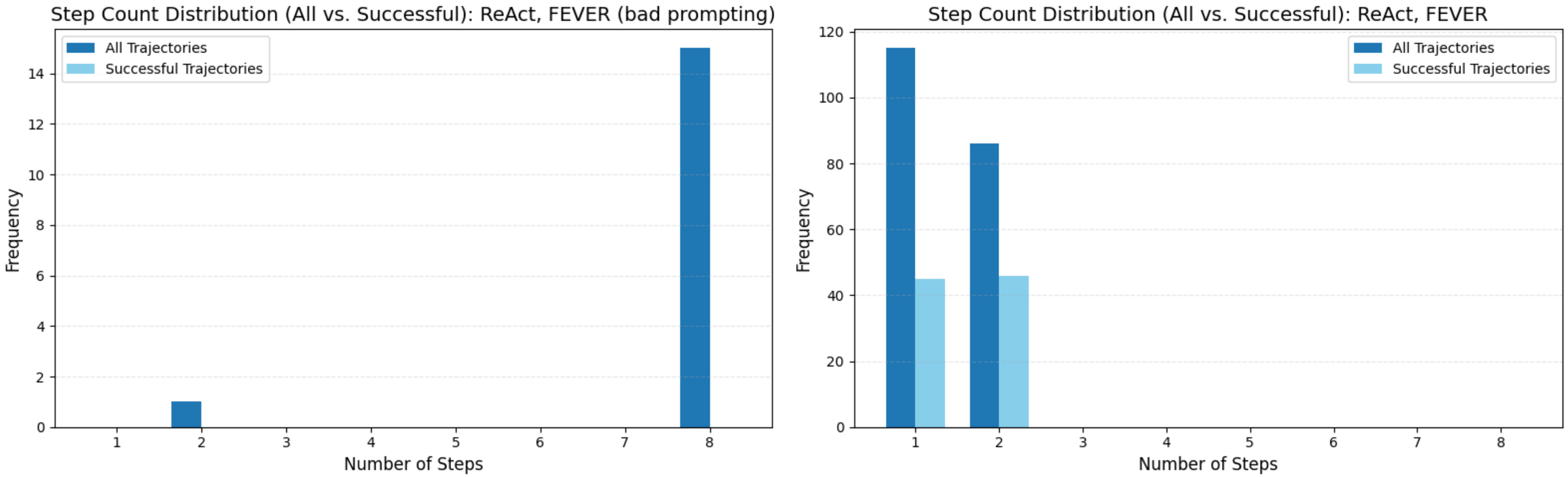
HotpotQA Benchmark



- Three-step trajectory results in longer answer (?)

FEVER Benchmark

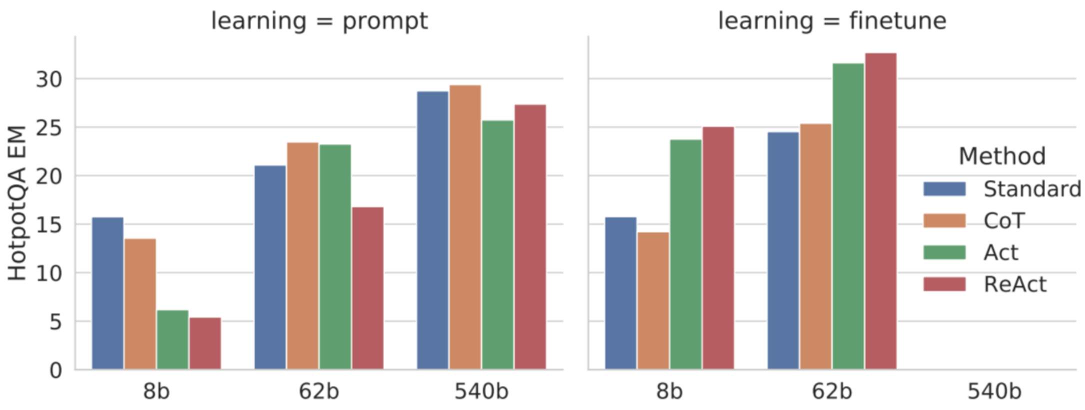
- Without proper prompting, agent becomes trapped in iterative loops of unproductive reasoning or action sequences.

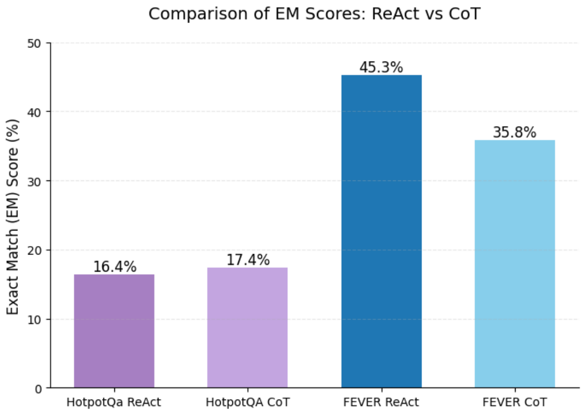


- Solution: add to context a message to follow strict pattern

(e. g. “In the last line of your answer write only in format: Action i: Finish[], nothing else.”)

Conclusions





In summary:

- Reproduced results align with the original authors' findings.
- ReAct demonstrates superior performance over Chain-of-Thought (CoT) when given optimal prompting or fine-tuning.
- Agent performance scales with its ability to learn the correct environment interaction format.

Thank you!

All results and code are available on
Github: https://github.com/olgafilimonova2004/ReAct_experiments